Customer Segmentation and Churn Pattern Analytics in European Banking

An exploratory study of customer attrition using an interactive analytical dashboard

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Tools: Microsoft Excel / Google Sheets (data preparation and analysis); Python and Streamlit (interactive dashboard)

Abstract

Customer attrition, commonly termed churn, represents a persistent challenge for retail banks, since the loss of existing customers reduces future revenue and increases the cost of acquiring replacements. This study examines churn patterns in a dataset of 10,000 European banking customers and compares attrition across geography, age, gender, product holding and membership activity. The overall churn rate was 20.37%, with 2,037 customers exiting and 7,963 retained. Notably higher rates were observed among customers in Germany (32.44%), those aged 46–60 (51.12%), female customers (25.06%) and inactive members (26.85%). Very high rates were also recorded for customers holding three or four products; however, these results require verification of group sizes before interpretation. The findings are descriptive rather than causal, and an interactive Streamlit dashboard was developed to support further exploration of the data.

Keywords: customer churn, customer segmentation, retail banking, exploratory data analysis, dashboard analytics

1. Introduction

Churn occurs when a customer discontinues the use of an organisation's services. In retail banking, such attrition carries both direct and indirect costs: future revenue is forfeited, and additional resources must be devoted to attracting new customers. Examining churn at the level of customer groups, rather than only in aggregate, allows an institution to identify where investigation and retention effort may be most worthwhile.

The present study explores churn patterns within a European banking dataset. Its purpose is twofold: first, to identify which customer segments exhibit comparatively higher attrition; and second, to present these results through an interactive dashboard that users can filter and inspect, rather than relying solely on static tables.

2. Objectives

The study pursued the following objectives:

1. To determine the number of customers who exited and the number who were retained.
2. To compare churn rates across geography, age, gender, product count and membership activity.
3. To develop an interactive dashboard incorporating summary indicators, charts and filters.
4. To use the results to identify areas for further investigation and possible retention initiatives.

3. Data and Preparation

3.1 Dataset

The dataset comprises 10,000 customer records. The variables used in the analysis are CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary and Exited, together with customer identifiers and a year field. The binary variable *Exited* indicates whether a customer left the bank.

3.2 Workbook structure

The project workbook was organised into separate tabs for the original data, cleaned data, validation, analysis, KPI summary and dashboard. The deployed Streamlit application requires the user to upload this workbook through the sidebar before any analysis is displayed.

3.3 Data handling

Customer identifiers and surnames are not required to explain the overall findings. Should the workbook be distributed beyond the project team, these fields ought to be handled with care or removed.

4. Methodology

The analysis began with the total customer count, which was divided into customers who exited and customers who were retained. The churn rate was calculated as the number of churned customers divided by the total number of customers, multiplied by 100. The same

calculation was then applied within individual customer groups so that their rates could be compared directly.

The following comparisons were undertaken:

- **Geography:** customers in France, Germany and Spain.
- **Age and gender:** differences in churn across demographic groups.
- **Membership activity:** active compared with inactive members.
- **Product count:** churn among customers holding different numbers of products.
- **Dashboard exploration:** use of filters to narrow the data and to examine the Overview, Customer Segments, Financial Profile and Data Explorer sections.

5. Results

5.1 Overall churn

Of the 10,000 customers in the dataset, 2,037 exited and 7,963 remained, yielding an overall churn rate of 20.37%. This figure serves as the baseline against which segment-level results are compared.

Table 1. Summary of churn results

Measure	Result
Total customers	10,000
Customers who exited	2,037
Customers retained	7,963
Overall churn rate	20.37%
Germany churn rate	32.44%
Age 46–60 churn rate	51.12%
Female customers	25.06%
Male customers	16.46%
Inactive members	26.85%
Active members	14.27%

Measure	Result
Customers with 3 products	82.71%
Customers with 4 products	100%

5.2 Segment-level observations

Several patterns emerge from the segment comparisons. Churn among customers in Germany (32.44%) is considerably higher than the overall rate of 20.37%. The 46–60 age group displays a notably high rate of 51.12%. Female customers (25.06%) churned at a higher rate than male customers (16.46%), and inactive members (26.85%) churned at a higher rate than active members (14.27%), which suggests that customer engagement merits closer examination.

5.3 Customers holding three or four products

The reported churn rates for customers holding three products (82.71%) and four products (100%) are exceptionally high. These figures should not be interpreted in isolation. The number of customers in each group has yet to be verified, and a small group can produce a very high percentage on the basis of only a few cases. As with all results in this study, the figures describe the data and do not establish why customers left.

6. Dashboard

An interactive dashboard was built in Streamlit to consolidate the principal indicators and charts. Users may filter results by geography, gender, age range, membership activity and customer status. The dashboard is organised into four sections:

- **Overview:** headline indicators and summary charts.
- **Customer Segments:** comparison of churn patterns across customer groups.
- **Financial Profile:** exploration of customers' financial characteristics.
- **Data Explorer:** inspection of the filtered data.

When the application is opened in a new session, the project Excel workbook must be uploaded through the sidebar, after which the customer totals and charts are displayed.

7. Discussion and Implications

The results suggest several directions for further work:

1. **Germany.** The segment should be reviewed in greater detail, with its size and customer value compared against other regions.
2. **Customers aged 46–60.** Further analysis is needed to establish whether attrition in this group relates to service needs, product suitability or other factors observable in the data.
3. **Inactive members.** Engagement initiatives could be trialled, with the effect on retention measured rather than assumed.
4. **Product count.** Group sizes and data quality should be confirmed before conclusions are drawn about customers holding three or four products.
5. **Longitudinal monitoring.** These measures should be tracked over time, since a single dataset provides a snapshot and cannot reveal whether a pattern is strengthening or weakening.

8. Limitations

This study is exploratory. It identifies differences between groups but cannot demonstrate cause and effect. Segment rates should be interpreted together with segment sizes. In addition, the available summary does not provide validated values for every potential indicator, such as revenue at risk or the proportion of high-value customers lost; these should be calculated and verified independently before being reported as final results.

Furthermore, the dataset represents a particular set of records, and the findings may not generalise to other banks or other periods. Any business action arising from this work should be tested and evaluated before wider implementation.

9. Conclusion

The analysis found an overall churn rate of 20.37% within a dataset of 10,000 customers. Segment comparisons indicate differences in attrition by geography, age, gender, product count and membership activity, and the accompanying dashboard allows these patterns to be explored through filters and charts. The most appropriate next steps are to validate segment sizes, to investigate the higher-risk groups in greater depth, and to measure the outcomes of any retention experiments.

Project Resources

- GitHub repository: european-bank-churn-analytics
- Live dashboard: European Banking Churn Analytics

Note: This document summarises the results available in the project analysis. Workbook values and segment counts should be confirmed before the findings are used for formal business decisions.